Gaussian processes regression

- Formulation
- GP regression
- Squared exponential kernel

GP regression

We have data $\left\{(x_n^\sharp, y_n^\sharp, \epsilon_n^\sharp)\right\}_{n=1}^{N^\sharp} \subset \mathbb{X} \times \mathbb{R} \times \mathbb{R}$ and seek an unknown function

$$f : \mathbb{X} \mapsto \mathbb{R}$$

Our regression model consists of

$$f \sim \mathrm{GP}_{\mathbb{X}}(m, v)$$
$$y_n^\sharp | f \sim \mathrm{Normal}\left(f(x_n^\sharp), (\epsilon_n^\sharp)^2\right), \qquad n = 1, \ldots, N^\sharp$$

Our model's "hyper-parameters" consist of

- a mean *...this is a function*

$$m : \mathbb{X} \mapsto \mathbb{R}$$

- a symmetric positive semi-definite kernel *...this is also a function*

$$v : \mathbb{X} \times \mathbb{X} \mapsto \mathbb{R}$$

GP posterior

Our regression model consists of

$$f \sim \mathrm{GP}_{\mathbb{X}}(m, v)$$

$$y_n^\sharp | f \sim \mathrm{Normal}\left(f(x_n^\sharp), (\epsilon_n^\sharp)^2\right), \qquad n = 1, \ldots, N^\sharp$$

- To obtain predictions at some $x^\star \in \mathbb{X}$

...we discretize the prior on $\{x_n^\sharp\}_{n=1}^{N^\sharp} \cup \{x^\star\}$

$$\mathbb{X}^{N^\sharp+1} \ni \begin{pmatrix} x_1^\sharp \\ x_2^\sharp \\ \vdots \\ x_{N^\sharp}^\sharp \\ x^\star \end{pmatrix} \equiv \begin{pmatrix} x^\sharp \\ x^\star \end{pmatrix} \in \mathbb{X}^{N^\sharp} \oplus \mathbb{X}^1$$

...and investigate the statistics of

$$\mathbb{R}^{N^\sharp+1} \ni \begin{pmatrix} f_1^\sharp \\ f_2^\sharp \\ \vdots \\ f_{N^\sharp}^\sharp \\ f^\star \end{pmatrix} \equiv \begin{pmatrix} f^\sharp \\ f^\star \end{pmatrix} \in \mathbb{R}^{N^\sharp} \oplus \mathbb{R}^1$$

GP posterior

Our original regression model consists of

$$f \sim \mathrm{GP}_{\mathbb{X}}(m, v)$$
$$y_n^\sharp | f \sim \mathrm{Normal}\left(f(x_n), \epsilon_n^2\right), \qquad n = 1, \ldots, N^\sharp$$

Our discretized regression model takes the form

$$\begin{pmatrix} f^\sharp \\ f^\star \end{pmatrix} \sim \mathrm{Normal}_{N^\sharp+1}\left(\begin{pmatrix} m^\sharp \\ m^\star \end{pmatrix}, \begin{bmatrix} V^{\sharp\sharp} & V^{\sharp\star} \\ V^{\star\sharp} & V^{\star\star} \end{bmatrix}\right)$$
$$Y^\sharp | f^\sharp \sim \mathrm{Normal}_{N^\sharp}\left(f^\sharp, P^{\sharp\sharp}\right)$$

Our discretized regression model decomposes like this

$$f^\sharp \sim \mathrm{Normal}_{N^\sharp}\left(m^\sharp, V^{\sharp\sharp}\right)$$
$$f^\star | f^\sharp \sim \mathrm{Normal}_1\left(m^\star + V^{\star\sharp}\left(V^{\sharp\sharp}\right)^{-1}\left(f^\sharp - m^\sharp\right), V^{\star\star} - V^{\star\sharp}\left(V^{\sharp\sharp}\right)^{-1} V^{\sharp\star}\right)$$
$$Y^\sharp | f^\sharp \sim \mathrm{Normal}_{N^\sharp}\left(f^\sharp, P^{\sharp\sharp}\right)$$

GP posterior

Our discretized regression model decomposes like this

$$
\begin{aligned}
f^\sharp &\sim \text{Normal}_{N\sharp}\left(m^\sharp, V^{\sharp\sharp}\right) \\
f^\star | f^\sharp &\sim \text{Normal}_1\left(m^\star + V^{\star\sharp}\left(V^{\sharp\sharp}\right)^{-1}\left(f^\sharp - m^\sharp\right), V^{\star\star} - V^{\star\sharp}\left(V^{\sharp\sharp}\right)^{-1} V^{\sharp\star}\right) \\
Y^\sharp | f^\sharp &\sim \text{Normal}_{N\sharp}\left(f^\sharp, P^{\sharp\sharp}\right)
\end{aligned}
$$

Our decomposed discretized regression model recomposes like this

$$
\begin{aligned}
f^\sharp &\sim \text{Normal}_{N\sharp}\left(m^\sharp, V^{\sharp\sharp}\right) \\
\begin{pmatrix} f^\star \\ Y^\sharp \end{pmatrix} \Bigg| f^\sharp &\sim \text{Normal}_{1+N\sharp}\left(\begin{pmatrix} m^\star + V^{\star\sharp}\left(V^{\sharp\sharp}\right)^{-1}\left(f^\sharp - m^\sharp\right) \\ f^\sharp \end{pmatrix}, \begin{bmatrix} V^{\star\star} - V^{\star\sharp}\left(V^{\sharp\sharp}\right)^{-1} V^{\sharp\star} & 0^{\star\sharp} \\ 0^{\star\sharp} & P^{\sharp\sharp} \end{bmatrix}\right)
\end{aligned}
$$

GP posterior

Our decomposed discretized regression model recomposes like this

$$f^\sharp \sim \text{Normal}_{N^\sharp}\left(m^\sharp, V^{\sharp\sharp}\right)$$

$$\begin{pmatrix} f^\star \\ Y^\sharp \end{pmatrix} \middle| f^\sharp \sim \text{Normal}_{1+N^\sharp}\left(\begin{pmatrix} m^\star + V^{\star\sharp}\left(V^{\sharp\sharp}\right)^{-1}\left(f^\sharp - m^\sharp\right) \\ f^\sharp \end{pmatrix}, \begin{bmatrix} V^{\star\star} - V^{\star\sharp}\left(V^{\sharp\sharp}\right)^{-1}V^{\sharp\star} & 0^{\star\sharp} \\ 0^{\star\sharp} & P^{\sharp\sharp} \end{bmatrix}\right)$$

Our recomposed discretized regression model marginalizes like this

$$\begin{pmatrix} f^\star \\ Y^\sharp \end{pmatrix} \sim \text{Normal}_{1+N^\sharp}\left(\begin{pmatrix} m^\star \\ m^\sharp \end{pmatrix}, \begin{bmatrix} V^{\star\star} & V^{\star\sharp} \\ V^{\star\sharp} & P^{\sharp\sharp} + V^{\sharp\sharp} \end{bmatrix}\right)$$

Our marginalized recomposed discretized regression model conditions like this

$$f^\star | Y^\sharp \sim \text{Normal}_1\left(m^\star + V^{\star\sharp}\left(P^{\sharp\sharp} + V^{\sharp\sharp}\right)^{-1}\left(Y^\sharp - m^\sharp\right), V^{\star\star} - V^{\star\sharp}\left(P^{\sharp\sharp} + V^{\sharp\sharp}\right)^{-1}V^{\sharp\star}\right)$$

GP posterior

We have data $\left\{(x_n^\sharp, y_n^\sharp, \epsilon_n^\sharp)\right\}_{n=1}^{N^\sharp} \subset \mathbb{X} \times \mathbb{R} \times \mathbb{R}$ and seek an unknown function

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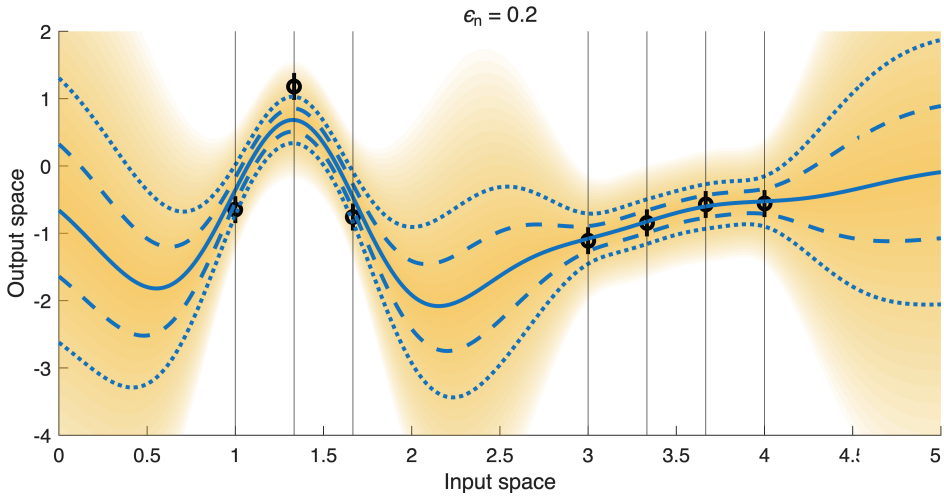
- Our regression model consists of

$$f \sim \mathrm{GP}_{\mathbb{X}}(m, v)$$
$$y_n^\sharp | f \sim \mathrm{Normal}\left(f(x_n^\sharp), (\epsilon_n^\sharp)^2\right), \qquad n = 1, \ldots, N^\sharp$$

- To obtain predictions at some $x^\star \in \mathbb{X}$, we apply the predictive

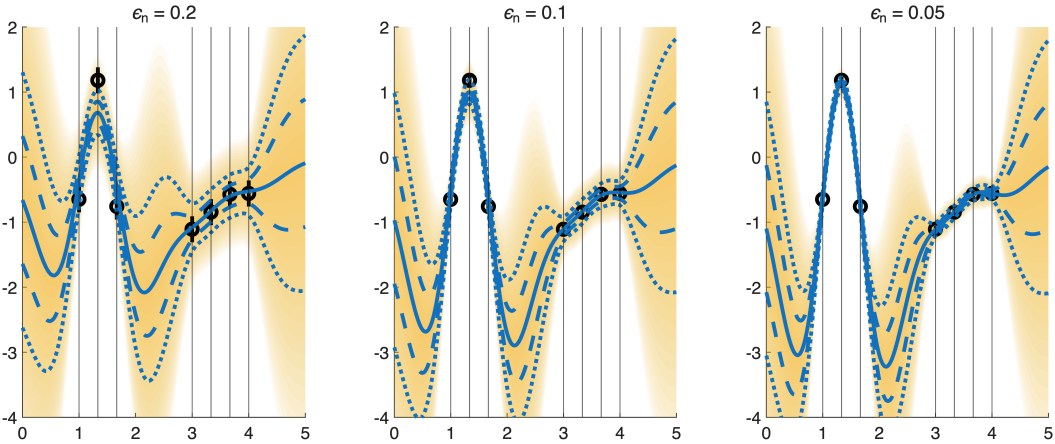
$$f^\star | Y^\sharp \sim \mathrm{Normal}_1\left(\mu^\star, (\sigma^\star)^2\right) \qquad f^\star \equiv f(x^\star)$$
$$\mu^\star = m^\star + V^{\star\sharp}\left(P^{\sharp\sharp} + V^{\sharp\sharp}\right)^{-1}\left(Y^\sharp - m^\sharp\right) \qquad \mu^\star \equiv \mu(x^\star)$$
$$\sigma^\star = \sqrt{V^{\star\star} - V^{\star\sharp}\left(P^{\sharp\sharp} + V^{\sharp\sharp}\right)^{-1} V^{\sharp\star}} \qquad \sigma^\star \equiv \sigma(x^\star)$$

Example



We have scalar inputs $\mathbb{X} = \mathbb{R}$ and $N^\sharp = 7$ points with $m(x) = 0$ and $v(x', x'') = \exp\left(-2\left(x' - x''\right)^2\right)$

Example



We have scalar inputs $\mathbb{X} = \mathbb{R}$ and $N^\sharp = 7$ points with $m(x) = 0$ and $v(x', x'') = \exp\left(-2\,(x' - x'')^2\right)$

The squared exponential kernel

The square exponential kernel is given by *...this is symmetric and positive semi-definite* $v : \mathbb{R} \times \mathbb{R} \mapsto \mathbb{R}$

$$v(x', x'') = \sigma^2 \exp\left(-\frac{1}{2}\left(\frac{x' - x''}{\ell}\right)^2\right) \qquad x', x'' \in \mathbb{R}$$

The kernel parameters consist of a strength σ and a scale ℓ

We consider a Gaussian process *...for some mean* $m : \mathbb{R} \mapsto \mathbb{R}$

$$f \sim \mathrm{GP}_{\mathbb{R}}(m, v)$$

and two inputs $x', x'' \in \mathbb{R}$

Our prior is

$$\begin{pmatrix} f(x') \\ f(x'') \end{pmatrix} \sim \mathrm{Normal}_2\left(\begin{pmatrix} m(x') \\ m(x'') \end{pmatrix}, \begin{bmatrix} v(x', x') & v(x', x'') \\ v(x'', x') & (x'', x'') \end{bmatrix}\right)$$

The squared exponential kernel

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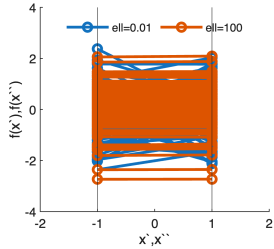
Our correlation is given by

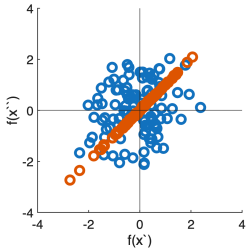
$$C(x', x'') = \frac{v(x', x'')}{\sqrt{v(x', x')}\sqrt{v(x'', x'')}} = \exp\left(-\frac{1}{2}\left(\frac{x' - x''}{\ell}\right)^2\right)$$

The limiting correlation is

$$\left|\frac{x' - x''}{\ell}\right| \gg 1 \qquad C(x', x'') \to 0$$

$$\left|\frac{x' - x''}{\ell}\right| \ll 1 \qquad C(x', x'') \to 1$$





The squared exponential kernel

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